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Systematic Screening of Optimal Signal Peptides for Secretory Production of Heterologous Proteins in *Bacillus subtilis*

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Supporting Information

ABSTRACT: *Bacillus subtilis* is widely used for large-scale industrial production of heterologous proteins. Because of its high intrinsic secretory capacity, it can efficiently secrete proteins into the culture supernatant via the general Sec-type secretion pathway. In this study, the α-amylase AmyS was used as a reporter to construct a library encompassing 173 Sec-type signal peptides (SPs) from *B. subtilis* using a fast, sequence-independent method. The resulting library DNA which harbored different signal peptides in the expression vector was used to transform *B. subtilis* directly at high efficiency, and 15 SPs which produced a significantly increased yield of AmyS were identified using a starch-iodine-based high-throughput assay. Furthermore, the correlation between the sequences of the best-performing signal peptides and their secretion efficiency was analyzed, which revealed several common properties of these SPs. Finally, high-cell-density fermentation of the α-amylase-producing strain with the best-performing signal peptide yielded a maximum of 5086 U/mL amylase at 66 h with a high productivity of 77.1 U/mL·h.

KEYWORDS: *Bacillus subtilis, signal peptide, high-throughput screening, α-amylase*

INTRODUCTION

Recombinant protein production in microbial cells is an important aspect of the modern biotechnological industry. Although intracellular expression of heterologous proteins in host cells is widely admitted and performed, recent achievements in secretory expression of recombinant proteins have encouraged both the scientific and industrial communities to explore the potency of using bacteria with a secretory ability for protein production. By using these secretory-type host cells, synthesized target proteins could be directly secreted and accumulated in the extracellular medium, which provides a cost-effective downstream purification processing. Besides, secretory expression of recombinant proteins prevents severe accumulation of heterologous proteins within host cells, which often results in cell toxicity and incorrect protein folding.[1,2]

Bacillus subtilis is a Gram-positive model bacterium which is widely used for industrial production of recombinant proteins such as alpha-amylase, protease, lipase, and other industrial enzymes. Because of the ability of the bacteria to secrete large amounts of a target protein into the culture medium and the availability of a low-cost downstream production and purification process, over 60% of commercial industrial enzymes are produced in *Bacillus subtilis* and relative *Bacillus* species.[3] In contrast to the frequently used recombinant protein expression host *Escherichia coli*, *Bacillus subtilis* has no risk of endotoxin contamination and has been certificated as a GRAS (generally regarded as safe) organism by the FDA, which makes it a better choice for food-grade and pharmatheutical protein production. The complete genome sequence of the *B. subtilis* model strain 168 is also available, which greatly promoted the construction of improved production strains using different genetic engineering tools.[4]

While recent findings have shown that *E. coli* is capable of secreting target proteins, heterologous proteins leading with a N-Terminal signal peptide can be secreted to the periplasm and eventually leaked and accumulated into the culture medium. Recent studies reveal that the extracellular leaking expression of heterologous proteins in *E. coli* may be caused by the hyperproduction of the target proteins,[5] and because of the absence of native secretory pathway in *E. coli*, its secretory production is often extremely low when compared with other native secretory-type expression hosts like *B. subtilis*. Therefore, *B. subtilis* is often a better choice for industrial-scale recombinant protein production. In *B. subtilis*, the export of protein is generally accomplished by the Sec-type secretion pathway, which governs over 90 percent of the secretory proteins found in *B. subtilis*. The N-terminal sequence of a secreted protein carries a specific secretion signal known as signal peptide. After the nascent peptide with the signal peptide was synthesized, it can be recognized and translocated by the components of the Sec-type secretory pathway through the membrane into the extracellular medium. Therefore, the signal peptide is definitely a key factor that determines the best pathway for the target protein and how it is secreted across the membrane.[6−8] The Sec-type signal peptides often consist of

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Table 1. Bacterial Strains and Plasmids Used in This Study

name	description	source
	Strains	
1A751p	*B. subtilis*, his apr npr eglS(DELTA)102 bglT/bglS (DELTA)EV, amyE::prsA	lab stock
1A751pSPwt	1A751-prsA harboring pSPwt vector	this study
1A751pSP1	1A751-prsA harboring pSP1 vector	this study
1A751pSP2	1A751-prsA harboring pSP2 vector	this study
1A751pSP3	1A751-prsA harboring pSP3 vector	this study
1A751pSP4	1A751-prsA harboring pSP4 vector	this study
1A751pSP5	1A751-prsA harboring pSP5 vector	this study
1A751pSP6	1A751-prsA harboring pSP6 vector	this study
1A751pSP7	1A751-prsA harboring pSP7 vector	this study
1A751pSP8	1A751-prsA harboring pSP8 vector	this study
1A751pSP9	1A751-prsA harboring pSP9 vector	this study
1A751pSP10	1A751-prsA harboring pSP10 vector	this study
1A751pSP11	1A751-prsA harboring pSP11 vector	this study
1A751pSP12	1A751-prsA harboring pSP12 vector	this study
1A751pSP13	1A751-prsA harboring pSP13 vector	this study
1A751pSP14	1A751-prsA harboring pSP14 vector	this study
1A751pSP15	1A751-prsA harboring pSP15 vector	this study

name	description	source
	Plasmids	
pMA5	*Bacillus* expression vector, *Hpa*II promoter, RepB, Km^r	lab stock
pMA5-AmyS	pMA5 vector containing the alpha-amylase gene *amyS* from *B. stearothermophilus*	this study
pMA5-PaprE-AmyS	pMA5-AmyS vector in which the *Hpa*II promoter was replaced by the aprE promoter	this study
pSPwt	pMA5-PaprE-Amys derivative, with amyS signal peptide	this study
pSP1	pMA5-PaprE-Amys derivative, with yvce signal peptide	this study
pSP2	pMA5-PaprE-Amys derivative, with yoqm signal peptide	this study
pSP3	pMA5-PaprE-Amys derivative, with yuab signal peptide	this study
pSP4	pMA5-PaprE-Amys derivative, with pel signal peptide	this study
pSP5	pMA5-PaprE-Amys derivative, with yoaw signal peptide	this study
pSP6	pMA5-PaprE-Amys derivative, with yqxi signal peptide	this study
pSP7	pMA5-PaprE-Amys derivative, with pelb signal peptide	this study
pSP8	pMA5-PaprE-Amys derivative, with lipa signal peptide	this study
pSP9	pMA5-PaprE-Amys derivative, with lipb signal peptide	this study
pSP10	pMA5-PaprE-Amys derivative, with yoqh signal peptide	this study
pSP11	pMA5-PaprE-Amys derivative, with sacb signal peptide	this study
pSP12	pMA5-PaprE-Amys derivative, with ybfo signal peptide	this study
pSP13	pMA5-PaprE-Amys derivative, with bgls signal peptide	this study
pSP14	pMA5-PaprE-Amys derivative, with yddt signal peptide	this study
pSP15	pMA5-PaprE-Amys derivative, with yobb signal peptide	this study

three specific domains: a positively charged N-domain, a hydrophobic H-domain, and a C-domain containing the signal peptidase digestion motif.[9] For efficient secretory production of recombinant proteins in *B. subtilis*, targeting secreted protein to the secretory machinery, the translocation and release process, and precise processing by signal peptidase (SPase) and signal peptide peptidase (SPP), these major bottlenecks in a secreted pathway should be addressed as a first priority.[10] This means the optimal choice of the signal peptide is always the first and most important factor to be considered. In a previous study, Chen et al. reported that the production of alpha-amylase AmyL from *B. licheniformis* and AmyS from *B. amyloliquefaciens* in *B. subtilis* reached 1352 and 2300 U/mL after promoter and signal peptide optimization.[6,11] Because of our limited knowledge of the characteristics of signal peptides that are required for efficient protein secretion, the in silico prediction of the optimum signal peptide for a specific target protein is not feasible to date.[12] Notwithstanding, the optimal signal peptide sequence can be identified by construction and screening of a high-capacity signal peptide library which the target protein was leading by different signal peptides in the expression vector. Combined with a suitable high-throughput screening method, this strategy has been proved feasible in optimizing secretory efficiency for many heterologous proteins. Watanabe et al.[13] constructed and screened 108 signal peptides derived from *Corynebacterium glutamicum* R using the alpha-amylase from *Geobacillus stearothermophilus* as the reporter protein. As a result, 11 optimal signal peptides were identified and shown a 50−150 times secretory amylase activity improvement than the classical corynebacterial secretory protein PS2 signal peptide. Brockmeier et al.[12] and Degering et al.[14] screened hundreds of in silico predicted Sec-type signal peptides from *B. subtilis* and *B. licheniformis* for optimizing amylase and protease secretory expression level. These studies revealed that the secretory expression level of target protein can be greatly affected by different signal peptides, and the optimum signal peptide for one recombinant protein was not always the best choice for another target protein, eventually was not compatible with the new target protein. This phenomenon was also proved by a previous study that a library of 76 Sec-type signal peptides was constructed and screened for high secretory production of two recombinant proteins (staphylococcal nuclease NucA and lactobacterial amylase AmyA) in *Lactobacillus plantarum*.[15] Because of the great enhancement of the secretory expression level by optimized signal peptides, constructing and evaluating a signal peptide library should be a reasonable and feasible way to improve the secretion efficiency of target proteins. However, the signal peptide library construction strategy used in these approaches has a low efficiency and is therefore time-

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consuming and costly, involving either ligating every signal peptide separately to ensure library integrity or cloning a mixture of all PCR-amplified signal peptides by introducing a unique restriction endonuclease site. It is also challenging to choose an appropriate RE-site, and the resulting ligation products may not be transformed into *Bacillus subtilis* efficiently due to the low transformation rate of *Bacillus* sp. ($\sim 10^3-10^4$ transformants per μg of DNA), which limits the effective library-generation capacity and follow-up screening work. Thus, developing a strategy for fast and efficient signal peptide library construction and transformation in *Bacillus subtilis* is of utmost importance. The low transformation rate may be due to the different DNA uptake process during the competence period of *B. subtilis*. Early research found that linearized multimeric plasmids are much more effective in DNA transformation than circular forms.[16,17] Mottes et al. also found that multimeric plasmids can be directly transformed into *Bacillus subtilis* with very high efficiency.[18]

Here we present a novel strategy to improve the extracellular production of alpha-amylase (AmyS) from *Geobacillus stearothermophilus* by generating and screening a high-capacity signal peptide library in *Bacillus subtilis* easily and effectively. A total of 173 Sec-type signal peptides from *B. subtilis* were fused to the target protein by a sequence-independent, PCR-based cloning method without using a restriction endonuclease or ligase.[19] The resulting multimeric plasmid library DNA harboring different signal peptides was transformed into *B. subtilis* efficiently, and it was screened for constructs with high extracellular alpha-amylase activity by a simple starch-iodine based high-throughput method. Furthermore, signal peptides optimized for the secretory expression of AmyS were identified and validated by high-density fermentation. Finally, the potential correlation between the optimal signal peptides' sequence constitution and their secretion capacity was also explored.

MATERIALS AND METHODS

Bacterial Strains and Plasmids. The bacterial strains and plasmids used in this study are listed in Table 1. The primers used to construct the amylase expression vector are listed in Table 3, and those used to amplify the 173 Sec-type signal peptides from *B. subtilis* 168 are provided in Table S1. *B. subtilis* strain 1A751p, which overexpresses the PrsA foldase gene, was used as an expression host and for cloning and plasmid preparation.[11] All DNA manipulations followed standard procedures.[20]

Media, Growth Conditions, and Preparation of Culture Supernatants. *E. coli* strains were grown in Luria−Bertani (LB) broth (5 g/L yeast extract, 10 g/L tryptone, 10 g/L NaCl, pH 7) at 37 °C with constant shaking at 200 rpm. *Bacillus subtilis* was cultivated in SR (super-rich) medium (25 g/L yeast extract, 15g/L tryptone, 3g/L K_2HPO_4, pH 7.2) at 37 °C supplemented with 50 mg/L kanamycin or 5 mg/L chloramphenicol. Amylase production was monitored on agar plates containing 1% (w/v) soluble starch (Solarbio, China). For screening, the cultures that inoculated from single colony were grown for 20 h at 37 °C in 96-deep-well plates (Greiner, Germany) using a high-throughput microplate shaker (600 rpm, 80% relative humidity; Minitron-INFORS; Biogen, Spain). The plates were sealed with silicone membrane (Solarbio, China) to prevent evaporation. Culture supernatants were collected by centrifuging at 4 °C, 3200*g* for 20 min, and samples were used immediately to determine alpha-amylase activity.

Transformation of *Bacillus subtilis*. *B. subtilis* was transformed by the two-step transformation method as described by Spizizen et al.[21] with minor modifications as follows: The *B. subtilis* culture was growth in Growth-Medium II until it reached a cell density of OD_{600} = 1. Subsequently, the culture was divided into 1 mL aliquots, multimeric plasmid DNA (1 μg DNA solution per 1 mL aliquot) was added, and the samples were incubated for 1−3 h at 37 °C. Transformants were selected on LB-agar plates with suitable antibiotics.

PCR Amplification of Signal Peptides. Signal peptide-encoding sequences of *B. subtilis* 168 were retrieved from the Signal Sequence Database (http://www.signalpeptide.de/index.php). A total of 173 signal peptide-encoding fragments from *B. subtilis* were amplified and used for library construction. AmyS with the wild-type signal peptide originating from *B. stearothermophilus* was used as the benchmark construct. Amplification of different signal peptide-encoding sequences was performed under standard PCR protocol in a 50 μL reaction volume, which contains 0.5 μmol of forward/reverse primer, 1 U of Q5 high fidelity polymerase (New England Biolabs, U.S.A.), and 50 ng of *Bacillus subtilis* genomic DNA. After gel electrophoresis, the amplified DNA was purified using the EZNA Cycle-Pure kit (OMEGA, U.S.A.) and quantified with Nanodrop 2000 (Thermofisher, U.S.A.).

Plasmid Construction and Cloning Strategy. The amylase gene from *B. stearothermophilus* was cloned between the *Nde*I and *Bam*HI sites of the pMA5 vector, yielding plasmid pMA5-amys. To achieve a higher amylase expression level, vector pMA5-PaprE-amys was constructed by replacing the original *hpaII* promoter from pMA5 with the *aprE* promoter from *B. subtilis* using a sequence-independent cloning method called "multimeric cloning" described by Chun You et al.[19] The inserted fragment encoding the *aprE* promoter was amplified from genomic DNA of *B. subtilis* 168 using the primers PaprE-F/R which contained 3′ and 5′ 25- to 30-bp overhangs homologous to vector sequences. The vector backbone was amplified from pMA5 using the primers pMA5-F2 and pMA5-R2 which contained 3′ and 5′ 25- to 30-bp overhangs homologous to the inset sequence. Finally, the desired expression vector pMA5-PaprE-amys was generated by fusing these DNA templates at a molar ratio of 1:1 by prolonged overlap extension PCR (POE−PCR) with Q5 polymerase, and the POE−PCR products (multimeric plasmid) were used to transform competent *Bacillus subtilis* cells directly, yielding the desired plasmid.

Construction of a 173 Sec-Type Signal Peptide Library in *Bacillus subtilis*. To generate the signal peptide library, signal peptides from *B. subtilis* were PCR amplified and cloned into the pMA5-PaprE-amys vector using the strategy described above with minor modifications as follows: Signal peptide-encoding sequences of *B. subtilis* 168 were retrieved from the Signal Sequence Database (http://www.signalpeptide.de/index.php). All 173 Sec-type signal peptide -encoding fragments predicted in the genome of *B. subtilis* were amplified with different primers which both contained 5′ and 3′ 20-bp overhangs homologous to the vector sequence. The vector backbone was amplified from pMA5-PaprE-amys using the primers pMA5-PaprE-amys-F1 and pMA5-PaprE-amys-R1. In order to guarantee a sufficient ligation efficiency for signal peptides with different lengths, the amplified signal peptide fragments were divided into three groups based on their DNA lengths (55−72 nt, 75−96 nt, 102−126 nt), and each was mixed with vector backbone at equimolar ratios, generating three AmyS expression vector sublibraries with different signal peptides using the multimeric cloning method described previously.[19] Finally, the three sublibraries were mixed together to yield the intact signal peptide library with all 173 different signal peptides from *B. subtilis*. AmyS with the native signal peptide originating from *B. stearothermophilus* was used as the benchmark construct.

High-Throughput Screening of Amylase High-Secretion Strains. The extracellular amylase activity was determined in agar plates and 96-well microplates. First, the transformants harboring different signal peptides were reinoculated and cultured on a trypan blue-starch agar plate at 37 °C for 16 h. The amylase activity was verified by observing the clear zones on the agar plate. Colonies with clear zones were transferred to individual wells of a 96-well microtiter plate with LB medium. To minimize evaporative loss during incubation, a silicone membrane (Solarbio, China) was used to cover the microplate. After they were cultured at 37 °C and 800 rpm

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for 21 h, the microplates were centrifuged at 4000*g* for 10 min to collect the supernatants. To prepare amylase assay samples, the collected supernatants were diluted 200 times with 0.1 M pH 6.0 phosphate-buffered saline (PBS). An aliquot comprising 40 μL of the resulting buffered amylase sample was mixed with 40 μL amylose (1 mg/mL; Solarbio, China) and incubated in a PCR heat block for 5 min at 70 °C. The reaction was then terminated by adding 20 μL of 1N acetic acid, followed by adding 100 μL iodine reagent (0.2% iodine and 2% potassium iodide). For the blank control, the amylase sample was replaced with an equal volume of 0.1 M phosphate buffer (pH 6.0). Following the color development, the absorbance at 570 nm was measured using a microplate reader (Bio-TEK Instruments, Winooski, VT, U.S.A.).

Scale-Up Fermentation and α-Amylase Activity Assay. The recombinant *B. subtilis* 1A751pSP4 harboring the amylase expression vector pMA5-PaprE-pel-amys was used to perform a 7.5-L scale-up fermentation with BioFlo 310 fermenter (New Brunswick Scientific, U.S.A.) utilizing a fed-batch strategy. Temperature was maintained at 37 °C, and the pH value was maintained at pH 7.0 by feeding 25% ammonia and 30% phosphoric acid. Aeration rate was controlled at 6 L/min, and the DO was maintained around 20% of the air saturation via a cascaded tuning of the stirring speed. Foam was controlled by supplement of the silicone-based antifoaming agent A8011 Antifoam C (Sigma-Aldrich, U.S.A.). To perform the fermentation, 8% of the seed inoculum was inoculated into the 7.5 L fermenter with 3.5 L of the initial fresh fermentation medium (3% tryptone (Oxoid, U.S.A.), 5% yeast extract (Oxoid, U.S.A.), and 0.6% K_2HPO4) supplemented with 50 μg/mL kanamycin (Solarbio, China) and 5 μg/mL chloramphenicol (Solarbio, China). When the cell growth rate reached the stable phase, feeding medium (5% yeast extract) was fed to the fermenter at a constant flow rate of 20 mL/h. The amylase activity was measured according to the 3,5-dinitrosalicylic acid (DNS) method.[22] One unit of α-amylase activity was defined as the amount of enzyme required to produce 1 μmol of glucose per min at 75 °C. All assays were carried out in duplicates. To determine the amylase activity, 80 μL of the substrate stock solution (1% (m/v) soluble starch (Solarbio, China) in 100 mM sodium acetate buffer, pH 6.0) was mixed with 40 μL of diluted enzyme solution. After the solution was incubated for 20 min at 75 °C, 200 μL of 1% DNS (Solarbio, China) was added to the solution, and the mixture was incubated in a boiling water bath for 10 min. After it was cooled to room temperature, the absorbance of the solution supernatant at 540 nm was measured using a SpectraMax M5 microplate reader (Molecular Devices LLC, San Jose, CA, U.S.A.). The enzyme-free blank sample was prepared in the same way as a negative control. All measurements were performed in triplicate.

SDS-PAGE Analysis. Culture samples were harvested and centrifuged at 12 000*g* and 4 °C for 10 min to collect the amylase-containing supernatant. After the SDS-PAGE loading buffer was added (Solarbio, China), the supernatants were incubated in a boiling water bath for 10 min and equal amounts of protein were loaded and separated on a NuPAGE 12% Bis−Tris Gel (Invitrogen Life Technologies, U.S.A.) with MOPS SDS Running Buffer (Invitrogen Life Technologies, U.S.A.) according to the manufacturer's instructions. The PageRuler Prestained Protein Ladder (Invitrogen Life Technologies, U.S.A.) was used as a molecular mass marker covering the 10- to 180-kDa range. Following electrophoresis, proteins were visualized by Coomassie Brilliant Blue R-250.

RESULTS

Design and Construction of a Library Encompassing All Sec-Type Signal Peptides of *B. subtilis*. In order to improve the secretory expression of heterologous amylase in *Bacillus subtilis* strains, a signal peptide library consisting of 173 Sec-type signal peptides was constructed. The alpha-amylase from *Geobacillus stearothermophilus* (AmyS) served as the target protein, and the *B. subtilis* strain 1A751p which overexpressed the *prsA* foldase gene was used as the expression host. In this study, signal peptide-encoding sequences of *B. subtilis* 168 were retrieved from the Signal Sequence Database (http://www.signalpeptide.de/index.php) to construct the signal peptide library. The length of the selected signal peptides varied from 19 to 42 residues. To minimize the differences of cloning efficiency of signal peptides with different lengths, the chosen signal peptides were first divided into three groups on the basis of their length range (Figure 1),

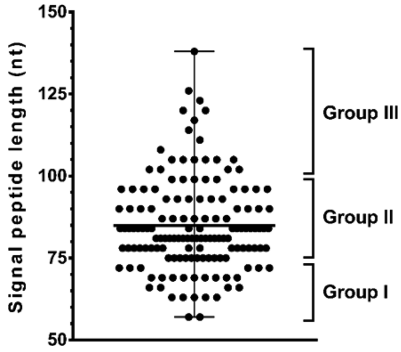


Figure 1. Length distribution of 173 Sec-type signal peptides from *B. subtilis* 168. To avoid differences of ligation efficiency, the signal peptides were divided into three groups: Group I (55−72nt), Group II (75−96nt), and Group III (102−126nt), and they were ligated separately at the same molar ratio to the vector backbone.

followed by generating three corresponding signal peptide sublibraries. Each sublibrary was constructed by fusing signal peptides in-frame to the mature AmyS amylase encoding sequence downstream of its constitutive *aprE* promoter in plasmid pMA5 using a "multimeric plasmid PCR cloning" strategy with minor modifications, as described in the Methods section. By using this cloning strategy (Figure 2), there is no need to introduce any restriction endonuclease sites between the signal peptide and target gene, which avoids the insertion of any additional residues into the mature target protein, or interference with signal peptide cleavage at the C-terminus of the signal peptide. Furthermore, a previous report[16] revealed that the Gram-positive bacterium *B. subtilis* prefers to uptake large, oligomeric plasmid DNA fragments during the competent period. Therefore, a multimeric plasmid containing different signal peptides could yield a high transformation rate in *B. subtilis* directly, without the need to employ a shuttle host. Finally, the multimeric plasmid DNA products of three sublibraries were combined, yielding the complete signal peptide-library DNA with all 173 different Sec-type signal peptides from *B. subtilis*.

To further guarantee the high transformation rate of the signal peptide library construct, factors which may affect the efficiency of the multimeric plasmid PCR cloning method, such as signal peptide-encoding fragment concentration and PCR cycle number, were also optimized (Figure 3). The results indicated that 2^{-5} to 1 ng/μL of the signal peptide DNA fragment, and 2^{-3} ng/μL achieved the highest transformation efficiency, yielding over 1400 CFU (colony-forming unit) on a 110 mm agar plate. Increasing or decreasing the fragment concentration drastically reduced the transformation rate in *B. subtilis*. In the test of PCR cycle numbers, the transformation efficiency gradually increased from 30 to 55 cycles, which implied that the tandem form of the multimeric plasmid DNA is preferentially taken up by the host cells. Finally, over 1800 colonies per plate were obtained using the

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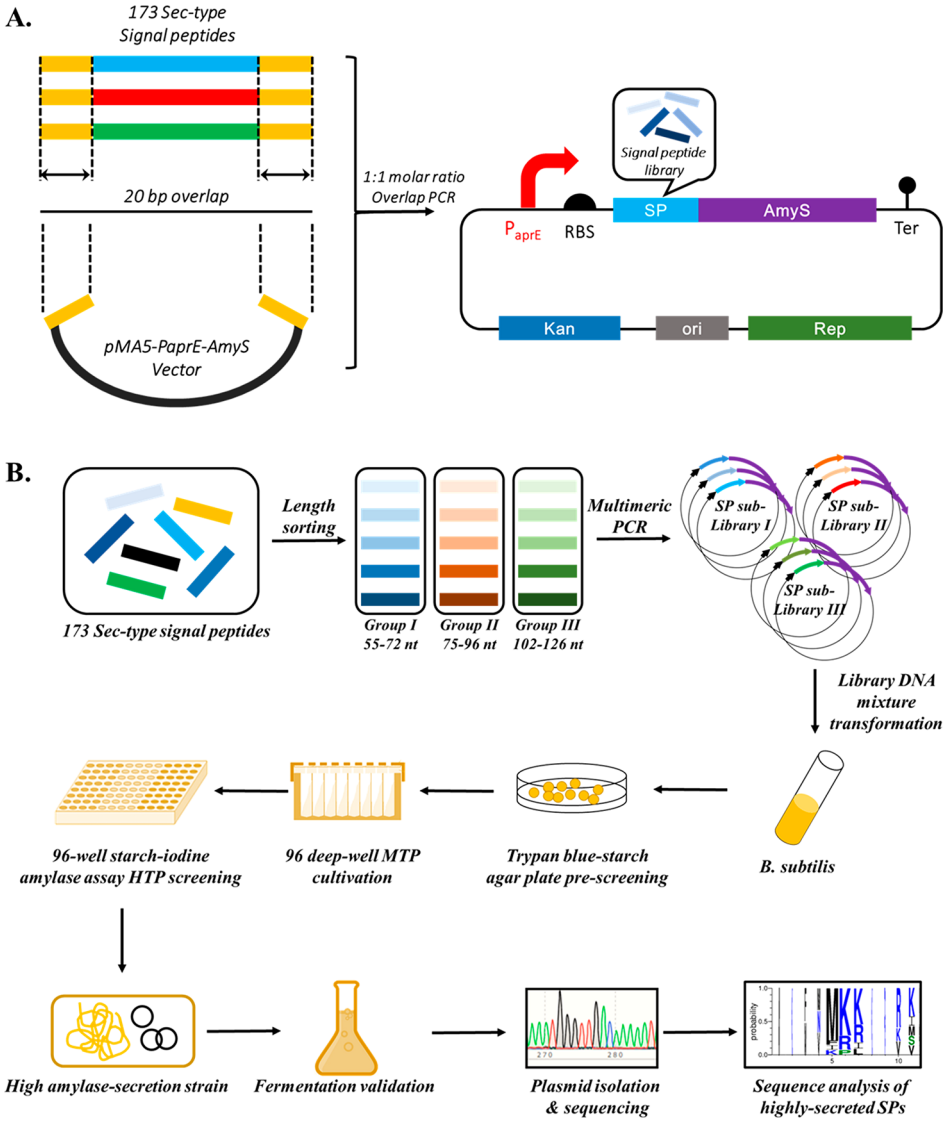


Figure 2. Cloning strategy and high-throughput screening workflow of the signal peptide library with 173 different signal peptides from *B. subtilis*. (A) Cloning strategy of 173 different signal peptides directly into the AmyS expression vector without introducing any RE sites or additional linker sequences. (B) Workflow of signal peptide-library construction and high-throughput screening.

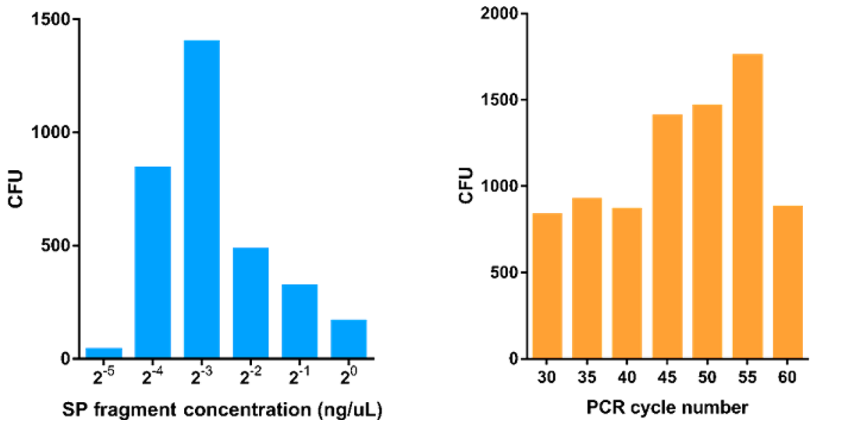


Figure 3. Optimization of the conditions of multimeric PCR for signal peptide-library construction. Transformation efficiency of DNA products obtained using different PCR conditions was determined by counting the colonies formed on the agar plate (CFU). The concentration of signal peptide fragments was optimized at a range of 2^{-5} to 1 ng/μL; the cycle number which determined the length of multimeric plasmid DNA was optimized at a range of 30 to 60 cycles.

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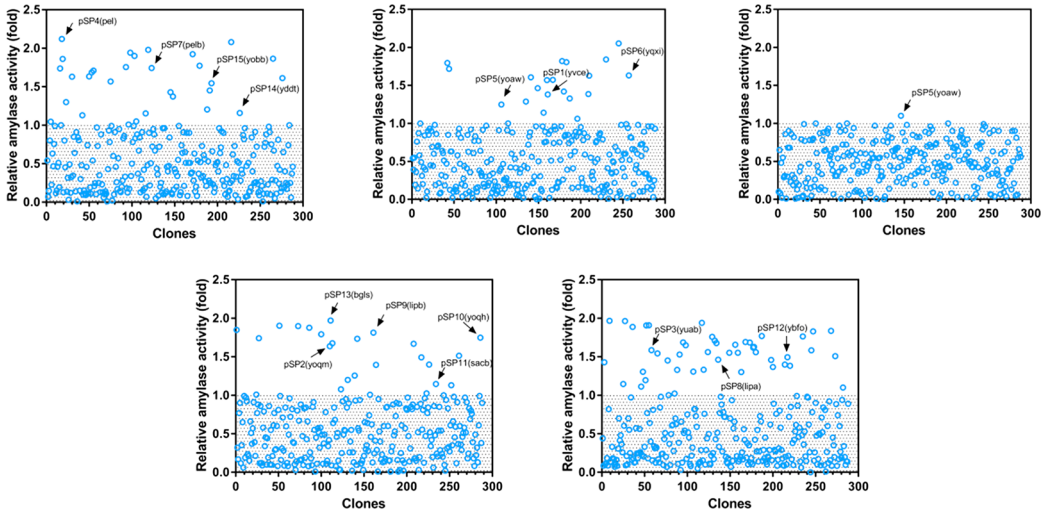


Figure 4. High-throughput screening of the secretory expression levels of amylase using a 96-well starch-iodine assay. Each open circle indicates the relative secretory amylase activity (number of folds) of a single clone picked from the trypan blue-starch prescreening agar plate.

Table 2. Information on the 15 Selected Best-Performing Signal Peptides for AmyS Amylase Secretion

name	length (aa)	N domain	H domain	C domain	D-score	pI	ΔG of folding (kcal/mol)
yvcE	30	MRKSL	ITLGLASVIGTSSFLI	PFTSKTASA	0.722	11.82	−35.5
yoqM	25	MKLRK	VLTGSVLSLGLL	VSASPAFA	0.539	11.82	−29.2
yuaB	30	MKRKL	LSSLAISALSLGLL	VSAPTASFAAE	0.872	10.79	−34.4
pel	21	MKKVM	LATALF	LGLTPAGANA	0.611	10.81	−28.0
pelB	24	MKRLC	LWFTVFSLFL	VLLPGKALG	0.589	10.79	−31.9
yoaW	24	MKKMLM	LAFTFLL	ALTIHVGEASA	0.554	9.54	−35.5
yqxI	28	MFKKL	LLATSALTFSLSLV	LPLDGHAKA	0.688	10.51	−32.0
lipA	32	MKFVKRR	IIALVTILMLSVTS	LFALQPSAKAA	0.797	12.55	−39.1
lipB	28	MKKVL	MAFIICLSLILS	VLAAPPSGAKA	0.656	10.51	−32.5
yoqH	23	MKRFI	LVLSFL	SIIVAYPIQTNA	0.431	10.45	−25.5
ybfO	28	MKRMIVR	MTLPLLIVCLA	FSSFSASARA	0.63	12.22	−35.1
sacB	29	MNIKKFAKQ	TVLTFTTALL	AGGATQAFA	0.527	11.1	−34.4
bglS	28	MPYLKR	VLLLLVTGLFMS	LFAVTATASA	0.668	10.45	−32.1
yddT	28	MRKKR	VITCVMAASLTLG	SLLPAGYASA	0.649	10.89	−37.1
yobB	25	MKIRK	ILLSSALSFGM	LISAVPALA	0.51	11.82	−40.2

optimized PCR cloning conditions, which could meet the demands of a library with 173 signal peptides. The resulting signal peptide-library DNA was then transformed and screened for extracellular amylase activity in *B. subtilis*. The AmyS expression vector with its native signal peptide originating from *B. stearothermophilus* was used as the benchmark construct.

High-Throughput Screening of α-Amylase Secretion Capacity. For screening purposes, the *B. subtilis* strain 1A751p which overexpressed the *prsA* foldase gene[11] was transformed with the signal peptide-library DNA. Over 1500 clones (8-fold oversampling) were obtained and then reinoculated onto agar plates containing trypan blue and starch, which can be used to visualize amylase activity. About 90% of the clones formed a clear halo around the colony, and these colonies were picked for cultivation in different microtiter plates. The strain harboring the expression vector with the native signal peptide of AmyS from *B. stearothermophilus* was used as the wild-type signal peptide control. To measure the secretion capacity of strains with different signal peptides in the library, cells were harvested at $OD_{600} \approx 1.8$ (late logarithmic phase). The culture supernatants were collected, diluted, and their amylase activities were determined by a starch-iodine based assay. Five rounds of screening were performed to cover the entire signal peptide library. As shown in Figure 4, a large variation in the secretion capacity was observed among the different clones. Importantly, only about 1% of the clones had no significant amylase activity in the culture supernatant, while the others yielded a continuum of activity levels spanning from 15% to 212% of the activity secreted using the native AmyS signal peptide. About 100 clones with improved secretion capacity over the native signal peptide of AmyS were chosen, and their plasmids were isolated and sequenced to determine the optimal signal peptide. Finally, 15 signal peptides (see in Table 2) were identified as candidates for improving the secretory expression of the alpha-amylase AmyS in *B. subtilis*.

To further validate the optimal effect of the chosen signal peptides on the α-amylase secretion level in *B. subtilis*, we performed shake-flask fermentations of the 15 constructs. Plasmids with optimal signal peptides were retransformed into *B. subtilis* strain 1A751p to exclude the influence of strain variation during the HTP screening procedure, which resulted in 15 denovo constructed recombinant strains. As shown in Figures 5 and S2, after 48 h of cultivation, and the extracellular α-amylase activity of the strains 1A751pSP1(yvce), 1A751pSP2(yoqm), 1A751pSP3(yuab), 1A751pSP4(pel), 1A751pSP5(yoaw), 1A751pSP6(yqxi), 1A751pSP7(pelb),

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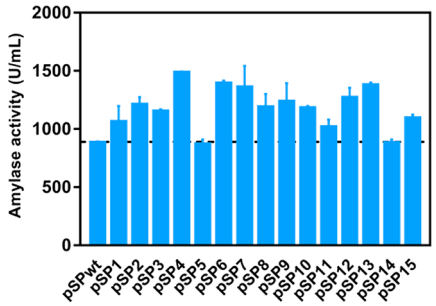


Figure 5. Amylase flask fermentation results of 15 recombinant strains with the screened best-performing signal peptides. The expression vector with the native signal peptide of AmyS is labeled as pSPwt, and the vectors carrying the 15 selected signal peptides after high-throughput screening are labeled as pSP1(yvce), pSP2(yoqm), pSP3(yuab), pSP4(pel), pSP5(yoaw), pSP6(yqxi), pSP7(pelb), pSP8(lipa), pSP9(lipb), pSP10(yoqh), pSP11(sacb), pSP12(ybfo), pSP13(bgls), pSP14(yddt), and pSP15(yobb). The extracellular amylase activity obtained with the native SP (pSPwt) of AmyS was defined as the benchmark.

1A751pSP8(lipa), 1A751pSP9(lipb), 1A751pSP10(yoqh), 1A751pSP11(sacb), 1A751pSP12(ybfo), 1A751pSP13(bgls), 1A751pSP14(yddt), 1A751pSP15(yobb) reached 1064.05, 1211.85, 1153.4, 1487.85, 871, 1394.15, 1360, 1191.6, 1238.25, 1181.75, 1019.1, 1270.1, 1379.55, 882.05, and 1096.2 U/mL, respectively (Table S2). Consistently with the HTP screening results, the extracellular α-amylase activity of the recombinant strain 1A751pSP4 was the highest, and it was about 68.4% higher than that of the strain with the native signal peptide. Thus, it could be concluded that the signal peptide pel is an excellent candidate for the secretory production of AmyS.

Correlation between Signal Peptide Properties and Secretion Efficiency. To explore the correlation between signal peptide sequence properties and secretion efficiency, we performed different analyses on the basis of the secretory efficiency (AmyS amylase activity in culture medium) and the properties of the 15 best-performing signal peptide sequences. D-scores and pI values of the native signal peptide of AmyS and the 15 signal peptides from *B. subtilis* were calculated using the SignalP 4.1 web server and the ExPASy Compute pI/Mw tool, and they were plotted against amylase activity to explore the potential correlation between them. However, no direct correlation was found between the secretory efficiency and D-score or pI value of these high-performing signal peptides (Figure 6A,C). Furthermore, given that the encoding sequences of some of the signal peptides possibly form a stable hairpin structure at the 5′ end of the amylase mRNA transcript, which may affect the translation initiation efficiency of the mRNA,[23−26] we also calculated the Gibbs free energies of folding of transcripts within a 170-nt fragment corresponding to the transcriptional initiation site (TIS) using the mfold program. This region of transcript interacts with the translation machinery earlier than the rest of the mRNA. As the predicted free folding energy profiles demonstrated (Figure 6B), the amylase activity of strains harboring the 15 signal peptides gradually increased compared to the native signal peptide of AmyS along with increasing folding free energy of the corresponding mRNAs. It implies that the expression levels of AmyS constructs with different signal peptides were affected by their 5′ terminal mRNA sequences, which may form a complex secondary structure that buries the Shine-Dalgarno (SD) sequences or initiator AUG region, resulting in a low initiation efficiency of the translation machinery.[24,27]

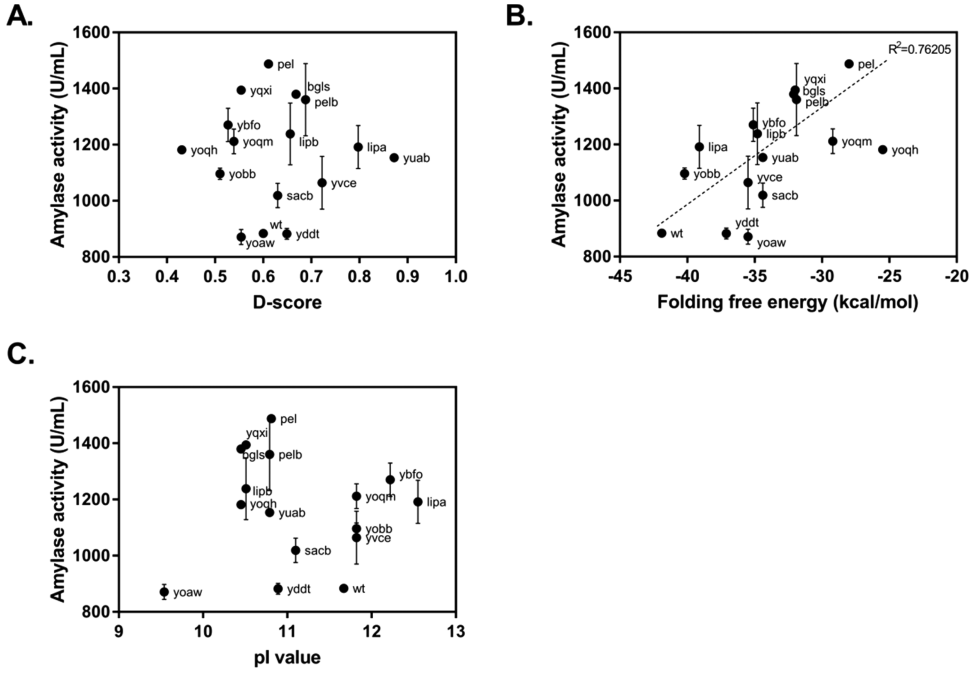


Figure 6. Correlation analysis between D-score, Gibbs free energy of folding, pI value, and amylase secretion efficiency of the 15 best-performing signal peptides and the wild-type signal peptide of AmyS. D-scores of different signal peptides were calculated using the SignalP 4.1 web server (http://www.cbs.dtu.dk/services/SignalP); the Gibbs free energies of folding of transcripts with different signal peptides were predicted using the mfold web server 3.6 (http://unafold.rna.albany.edu/?q=mfold/RNA-Folding-Form); the pI values of different signal peptides were predicted using the Compute pI/Mw tool on ExPASy (https://web.expasy.org/compute_pi/).

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Table 3. Primers Used in This Study[a]

primer	sequence (5′-3′)
AmyS-F	TATCATCATATGAAACAACAAAAACGGCTTTACGC
AmyS-R	TCTAGAGGATCCTTAAGGCCATGCCACCAGTCTAGGC
pMA5-F1	TGGCCTTAAGGATCCTCTAGAGTCGAGCTCAAGCTAG
pMA5-R1	TGTTGTTTCATATGTAAATCGCTCCTTTTTAGGTG
PaprE-F	ATAAAGTATAGTGTGTTATACTCTATTTTCGTTCTTTTCTGTATG
PaprE-R	TTTTGTTGTTTCATATGTCTTTACCCTCTCCTTTTAAAAAA
pMA5-F2	GAGGGTAAAGACATATGAAACAACAAAAACGG
pMA5-R2	AAAAGAACGAAAATAGAGTATAACACACTATACTTTATATTCATAAAGTG

[a]The *Nde*I and *Bam*HI site sequences are underlined.

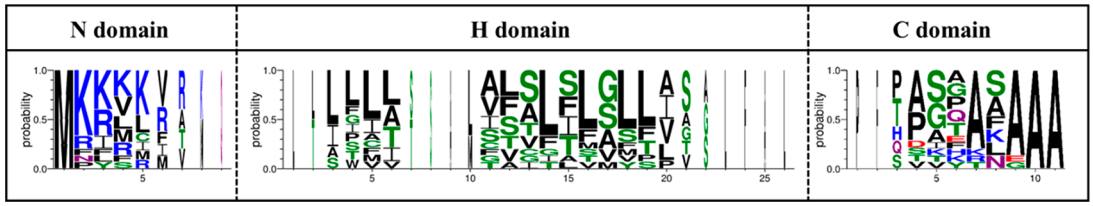


Figure 7. Sequence logos constructed from alignments of N-, H- and C-domains of the 15 best-performing signal peptides after HTP screening for AmyS secretion. The sequences were aligned using the MUSCLE algorithm of MEGA 5. Sequence logos were generated based on the alignments using WebLogo 3 (http://weblogo.threeplusone.com/create.cgi). Residues are colored according to their chemical features: polar (G, S, T, Y, C), green; neutral (Q, N), purple; basic (K, R, H), blue; acidic (D, E), red; hydrophobic (A, V, L, I,P, W, F, M), black). A probability value of 1.0 = 100%.

To analyze the potential correlation between high secretory efficiency and the amino acid composition of the best-performing signal peptides, 15 signal peptides were divided into N-, H- and C-domains using the Hidden-Markov model (HMM) in SignalP 2.0. The N-domain was defined to span from the start codon methionine to the last positively charged residue in the N-terminal part of the signal peptide. The rest of the signal peptide sequence was separated into an H-domain that is rich in hydrophobic amino acid, and a C-domain that ends with a predicted cleavage site of the signal peptidase (Table 2). Sequence alignment of the N-, H-, and C-domains of the 15 best-performing signal peptides was conducted using the MUSCLE algorithm in MEGA 5 software. On the basis of the alignment results, sequence logos of the different domains were generated using the WebLogo 3 web server. The alignment results did show some common properties (Figure 7): (1) a positively charged amino acid, lysine, occurs frequently at the second to fourth codon in the N-domain, especially at the second codon; (2) a highly hydrophobic H-domain arising from a stretch of leucine amino acids and a helix breaker residue, serine or glycine, scattered between the leucine codons in the middle; and (3) unsurprisingly, the consensus signal peptide cleavage motif A-X-A is particularly favored at the end of the C-domain, while an unbiased codon composition was found in front of the cleavage motif.

Fermentation of the α-Amylase-Producing Strain with Optimized Signal Peptide in 7.5-L Fermenter. Previously, Nathan et al. reported that A much higher productivity (9070 U mL^{-1}) of amylase was achieved using large-scale fermentation in *B. subtilis* WB800.[28] The extracellular expression capacity of *B. subtilis* 1A751pSP4 with the pel signal peptide was further explored in a 7.5-L fermenter. The fermenter was inoculated with 5% (v/v) of freshly cultured 1A751pSP4 (pMA5-PaprE-pel-amys) grown in SR medium at 37 °C for 18 h. To achieve a high cell density and α-amylase production, we performed the fermentation using a fed-batch strategy. Once the cell growth reached the steady state, a solution comprising 8.0% (w/v) soluble starch was added at a constant flow rate until the concentration in the fermenter reached 4% (w/v). As shown in Figure 8, the maximum OD_{600} in the fermenter reached 33.1 at 60 h. The activity of α-amylase in the medium continuously increased and reached a maximum of 5086 U/mL at 60 h with a high productivity of 77.1 U/mL·h (Table S3). Culture supernatant samples obtained at 24, 30, 42, 48, 54, 60, and 72 h were subjected to SDS-PAGE analysis to determine the secretory expression levels during the fermentation. As shown in Figure S1, the secreted α-amylase was detected at an apparent size of approximately 60 kDa, which was consistent with the theoretical molecular weight (58.8 kDa) of the mature AmyS protein. No amylase precursor band was detected on the gel, which indicates that the signal peptide pel led to the complete

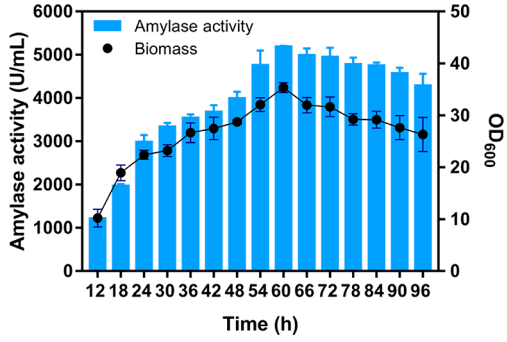


Figure 8. High-density fermentation results of the α-amylase AmyS recombinant strain 1A751pSP4 with the optimal signal peptide pel in a 7.5-L fermenter. To determine the α-amylase activity and the biomass, culture samples were collected every 6 h from 12 to 96 h. The solid squares indicate the biomass of the strain 1A751pSP4; solid circles indicate the α-amylase activity in the culture supernatant. The results represent the means ± SD from three independent experiments.

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processing of the AmyS protein precursor. The secretory expression level of the AmyS protein continuously increased from 24 h and achieved the highest level at 60 h, at which point it constituted up over 70% of the total secreted proteins in the culture supernatant, as calculated using QuantityOne software.

DISCUSSION

Bacillus subtilis is a Gram-positive bacterium which is widely used for large-scale industrial production of secreted recombinant proteins. Consequently, extensive research was conducted on the optimization of the secretory expression of heterologous proteins in *Bacillus subtilis*. Several studies addressed strategies to relieve bottlenecks for the secretion of heterologous proteins.[29,30] However, because the signal peptides from some proteins that are secreted by *B. subtilis* in large amounts, such as AprE and AmyE, are often not optimal for a given heterologous target protein,[31] the screening of signal peptide libraries in situ in the expression host seems pertinent. However, given the low transformation efficiency of *B. subtilis* cells, library DNA constructs could not be used directly. Consequently, cells with high DNA transformation efficiency such as *E. coli* were used as shuttle hosts to first construct the signal peptide-library, after which plasmids harboring different signal peptides were isolated as a DNA mixture followed by reintroduction into *B. subtilis* host cells.[12−15] Moreover, restriction endonuclease sites (RE sites) are widely used in signal peptide-library construction,[14,15,32,33] which introduces two additional residues into the target protein after signal peptidase cleavage. Thus, the workflow of the signal peptide-library construction was very time-consuming, and the additional residues introduced by the RE sites may influence the cleavage efficiency of the signal peptidase, resulting in low efficiency of screening for optimal signal peptides.

In this study, we used the α-amylase AmyS from *B. stearothermophilus* as a model protein and studied the influence of different Sec-type signal peptides on its secretion efficiency directly in *B. subtilis*. A library consisting of 173 different Sec-type signal peptides fused to the AmyS encoding sequence was constructed rapidly using an enzyme-free, RE site-free cloning strategy called "multimeric PCR cloning". The resulting signal peptide-library DNA was directly used to transform *B. subtilis* cells and was evaluated by a high-throughput activity screening method based on starch-iodine complexation in the culture medium. Finally, 15 best-performing signal peptides were selected, among which the pel signal peptide resulted in an almost 1.7-fold increase of amylase activity in the culture medium compared with the native signal peptide of AmyS. Interestingly, our screening did not identify some typical signal peptides such as those of AmyE or NprE, which were recognized to be secreted at high levels and were used widely for the secretory expression of heterologous protein. Instead, 9 out of the 15 best-performing signal peptides originated from uncharacterized so-called "Y" proteins. This corroborates the hypothesis that universal optimal signal peptides that are suitable for most heterologous proteins might not exist. Thus, signal peptide-library construction and high-throughput screening seem to be the only way to identify optimal signal peptides for heterologous proteins. Similar observations were also made in previous studies on optimizing the secretion of cutinase and esterase.[12,34,35] The authors found that the best signal peptide for cutinase secretion was a poor signal peptide for the secretion of the esterase and vice versa. These results clearly demonstrated that for different target proteins, a universal promising signal peptide can merely be identified, so individual optimum signal peptide for each target protein should be screened and identified.

To further explore the potential correlation between the properties of the best-performing signal peptides and their high secretion efficiency, we analyzed them with respect to their lengths, pI values, D-scores, and folding free energies of their corresponding mRNA structures. However, it seems that there is no direct correlation between the secretion efficiency of AmyS and the D-score or pI value of the signal peptides. However, the Gibbs free energies of folding of the transcripts encoding the 15 best-performing SPs revealed an obvious rising pattern along with the increase of the secretory expression level. Previously, Ng et al. also reported that a tight folding of the mRNA around the RBS yields low secretion of amylase via the signal peptide Usp45 in *Lactococcus lactis*, and secretion level gradually increased as the mRNA structure around the RBS was relaxed. We therefore speculated that the secretory expression levels of the amylase protein with different signal peptides were mostly affected by the mRNA secondary structure. In the relatively low-performing signal peptides, such as yobb or the native signal peptide of AmyS, a complex structure at the 5′ end of the mRNA transcript may bury the Shine-Dalgarno (SD) sequences or initiator AUG regions, resulting in a low translation initiation efficiency. By contrast, in the best-performing signal peptides such as pel, a more open mRNA secondary structure with a higher folding free energy in the 5′ UTR of the transcript was formed, which improved the access of the translation machinery. In further analysis, the sequences of these signal peptides were divided into N-, H-, and C-domains using the HMM model of the SignalP 2.0 webtool. The alignment results of N-, H-, and C-domains from 15 best-performing signal peptides showed some common properties. First, a net positively charged N-domain of signal peptides was observed and the amino acid lysine occurred frequently in this region, especially at the second codon downstream of the "ATG" start codon. The positive charges in the N-domain is required for targeting the signal peptide sequence into the inner membrane by electrostatic interactions between the loop-formed, negatively charged lipid head groups and the lipid layer, which play an important role in protein translocation process. Furthermore, the AAA codon of lysine may permit strong translation initiation of the target protein.[36] Second, a highly hydrophobic H-domain arising from a stretch of leucines was observed. A sufficient level of hydrophobicity is needed for efficient processing of the signal peptide and translocation of the protein. Rusch et al.[37] demonstrated that signal peptides less than five hydrophobic residues (such as leucine) in the H-domain exhibit very low secretory efficiency for the alkaline phosphatase. Additionally, the presence of a serine or glycine that breaks the congregating of the hydrophobic residues in the middle of the H-region may orient the signal peptide away from the SRP component and toward the Sec pathway, due to the decrease of the highly hydrophobic signals of the SRP. Third, the consensus signal peptide cleavage motif "A-X-A" was particularly favored at the end of the C-domain, which was consistent with the "−3, −1 rule". These observations in these high secretory efficiency signal peptides may indicate a new avenue for the rational optimization of signal peptides for the secretion of heterologous proteins, especially for alpha-amylase.

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■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jafc.8b04183.

SDS−PAGE analysis of the extracellular fraction of 1A751pSP4 harboring the optimized pel signal peptide; oligonucleotides used for PCR-amplification of 173 Sec-dependent signal peptides (SPs) from *B. subtilis* 168; alpha-amylase activity of 15 recombinant strains with the best-performing screened signal peptides after 36 h of flask fermentation; and alpha-amylase activity of the AmyS recombinant strain 1A751pSP4 with the best signal peptide pel in a 7.5-L fermenter (PDF)

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Notes

The authors declare no competing financial interest.

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